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Plausible Conjectures: Prospects for Explaining Several Long-standing Puzzles in Quantum Mechanics

Based on the framework described above, we can propose plausible conjectures and directions of explanation for several long-standing and unresolved puzzles in quantum mechanics:

1. The Ontological Foundation of State Vectors and Hilbert Space

In standard quantum mechanics, the state of a physical system is described by a vector in a complex Hilbert space, a mathematical structure regarded as a self-evident postulate. In our framework, the phase coordinates of wave packets on the spacetime lattice naturally constitute a complex representation space — phase corresponds to the argument, and the size of the wave packet to the modulus. Hilbert space is not an abstract mathematical convention, but a faithful mapping of the phase configurations within the World Phase. The Hermiticity of operators and spectral decomposition are then natural mathematical expressions of the periodic process of annihilation and rebirth.

2. The Measurement Problem and Wave Function Collapse

The measurement problem has long troubled the foundations of quantum mechanics: why does the act of observation cause the wave function to “collapse,” leaping from a superposition state to a single eigenstate? What is the mechanism of collapse? In our framework, collapse can be reinterpreted as a forced resonant synchronization of lattice states within the Electromagnetic Phase window. When a quantum system couples with a measuring apparatus, the two achieve resonant consensus within the Electromagnetic Phase window, and the lattice state of the system is “frozen” into an eigenstate consistent with the apparatus. Collapse is no longer a mysterious action at a distance, but a lattice synchronization event bounded by the Electromagnetic Phase window. The probabilistic nature of collapse can be traced back to the residual random fluctuations of the void breaking during the Chaotic Period — the probabilistic weights of the Born rule are essentially the statistical weights of the lattice resonance pathways.

3. The Intrinsic Origin of the Uncertainty Principle

The Heisenberg uncertainty principle is regarded in standard quantum mechanics as a fundamental principle-level limitation, but why does it exist? In our framework, the root of the uncertainty principle lies in the discreteness of the spacetime lattice. The Planck length and Planck time, as the minimum spatial step and temporal step of the lattice, imply that the position and momentum of a particle cannot be simultaneously defined

with infinite precision — every measurement is a “sampling” on the discrete lattice, and the granularity of the sampling naturally imposes an upper limit on precision. Uncertainty is not a limitation of knowledge, but a direct consequence of the discrete structure of spacetime.

4. Identical Particles and Quantum Statistics

In standard quantum mechanics, the indistinguishability of identical particles and the resulting Bose-Einstein and Fermi-Dirac statistics are empirical inputs. In our framework, the identical particle principle is already a core presupposition — all particles are essentially the same wave-packet entity, their differences arising solely from their phase positions within the World Phase. Exchange symmetry and antisymmetry can be directly derived from the phase relationships of wave packet rebirth in the three-valued resonance sequence: bosons correspond to constructive superposition in the same phase, and fermions to destructive constraints in opposite phases. Quantum statistics is thus no longer an empirical rule, but a combinatorial result of the phase logic of the lattice.

5. Quantum Entanglement and Non-Localicity

The instantaneous correlation exhibited by two entangled particles has long challenged classical intuition. In our framework, entanglement can be understood as two particles sharing the same borrowing-and-returning pair of the void breaking — they form a correlated phase relationship on the lattice at the moment of their birth, a relationship that does not depend on spatial distance but is encapsulated in the topological connections of the lattice. The non-local correlation verified by Bell’s inequality is not a “spooky action” of superluminal signal propagation, but a natural remnant of the non-locality of the void breaking at the particle level — after all, the first void breaking itself occurred simultaneously everywhere.

6. Wave-Particle Duality

The perplexity of wave-particle duality lies in the question: why does a single entity sometimes behave as a particle and sometimes as a wave? In our framework, this perplexity dissolves naturally. The propagation of a wave packet on the discrete lattice possesses both particle nature (each rebirth is an event occurring locally on a specific lattice) and wave nature (the phase relationships between adjacent lattices produce interference patterns). Duality is not a dual personality of an entity, but two complementary perspectives on the same discrete propagation process: particle nature is the local updating of the lattice, and wave nature is the phase correlation between lattices.

7. Quantum Tunneling Effect

Quantum tunneling — the phenomenon of a particle traversing a classically forbidden region — appears mysterious in the standard framework. In our framework, tunneling can be understood as the probabilistic jumping of the annihilation-rebirth mechanism between adjacent lattices. The classically forbidden region is merely an energy restriction within the Electromagnetic Phase window of the World Phase, while the residual random fluctuations of the void breaking allow the lattice state to be reborn with a finite probability in the “forbidden” region. The probability of tunneling depends on the degree of stability of the lattice resonance in that region.

8. Vacuum Fluctuations and the Casimir Effect

In standard quantum field theory, vacuum fluctuations are described as a process in which virtual particle pairs are constantly created and annihilated. In our framework, vacuum fluctuations correspond to the residual local breakings on the lattice that have not yet fully achieved resonant synchronization — these residues manifest as weak random fluctuations on the ordered substrate of the Stable Period. The attractive force between two parallel plates in the Casimir effect is then the net pressure difference created by the geometric restriction of lattice fluctuation modes between the plates: the space outside the plates can accommodate more fluctuation modes, thereby producing a net inward thrust.

Outlook

The above plausible conjectures demonstrate that the theoretical framework constructed in this paper holds the prospect of broadly explaining the core puzzles of quantum mechanics. Of course, substantial work remains to be done in progressing from qualitative conjecture to quantitative calculation — redefining field operators on the discrete lattice, deriving propagators, establishing Feynman rules, and verifying that the effective field theory form of standard quantum field theory is recovered in the long-wave limit will be the crucial test for whether this framework can truly gain a firm foothold.

Nevertheless, we have already taken the first step: starting from a simple question, relying on physical intuition and repeated deep contemplation, we have constructed a logically self-consistent picture of the universe rooted in first principles. The world should not be this complicated — and perhaps, it really is not as complex as we have imagined.

合理性猜想：对量子力学若干疑难的解释前景

基于上述框架，我们可以对量子力学中若干长期悬而未决的疑难提出合理性的猜想与解释方向：

1. 态矢量与希尔伯特空间的本体论基础

在标准量子力学中，物理系统的状态由复希尔伯特空间中的矢量描述，这一数学结构被视为不证自明的公设。在我们的框架中，波包在时空晶格上的相位坐标天然构成一个复数表示空间——相位对应于幅角，波包大小对应于模。希尔伯特空间不是抽象的数学约定，而是世界相内相位配置的忠实映射。算符的厄米性与谱分解，则是湮灭重生周期性过程的自然数学表达。

2. 测量问题与波函数坍缩

测量问题长期困扰着量子力学的基础：为何观测行为会导致波函数的“坍缩”，从叠加态跃迁至单一本征态？坍缩的机制是什么？在我们的框架中，坍缩可被重新理解为电磁相窗口内晶格状态的强制共振同步。当一个量子系统与测量仪器耦合时，两者在电磁相窗口内达到共振一致，系统的晶格状态被“定格”为与仪器一致的本征态。坍缩不再是神秘的超距作用，而是一场由电磁相窗口限定的晶格同步事件。坍缩的概率性则可追溯至虚空破缺在混沌期的残余随机涨落——玻恩规则的概率权重，实质上是晶格共振路径的统计权重。

3. 不确定性原理的内禀根源

海森堡不确定性原理在标准量子力学中被视为基本的原理性限制，但它为何存在？在我们的框架中，不确定性原理的根源在于时空晶格的离散性。普朗克长度和普朗克时间作为晶格的最小空间步长和时间步长，意味着粒子的位置和动量不可能被同时无限精确地定义——每一次测量都是一次在离散晶格上的“采样”，采样的粒度天然设置了精度的上限。不确定性不是认识的局限，而是时空离散结构的直接推论。

4. 全同粒子与量子统计

在标准量子力学中，全同粒子的不可区分性及由此产生的玻色-爱因斯坦统计和费米-狄拉克统计，是经验性的输入。在我们的框架中，全同粒子原则已是核心预设——所有粒子本质上是同一波包实体，其差异仅来源于在世界相中的相位位置。交换对称性与反对称性可由三值谐振序列中波包重生的相位关系直接导出：玻色子对应于同相位的叠加增强，费米子对应于反相位的相消约束。量子统计从此不再是经验规律，而是晶格相位逻辑的组合学结果。

5. 量子纠缠与非局域性

量子纠缠中两个粒子展现出的瞬时关联，长期以来挑战着经典直觉。在我们的框架中，纠缠可被理解为两个粒子共享同一虚空破缺的借还配对——它们在诞生之初便在

晶格上形成了关联的相位关系，这一关系不依赖于空间距离，而是封装在晶格的拓扑连接中。贝尔不等式所验证的非局域关联，并非信号超光速传播的“幽灵作用”，而是虚空破缺非局域性在粒子层面的自然遗存——毕竟，第一次虚空破缺本身就是全局同时发生的。

6. 波粒二象性

波粒二象性的困惑在于：一个实体为何时而表现为粒子，时而表现为波？在我们的框架中，这一困惑自然消解。波包在离散晶格上的传播既具有粒子性（每一次重生都是局域发生在特定晶格上的事件），又具有波动性（相邻晶格之间的相位关系产生干涉图样）。二象性不是实体的双重性格，而是同一离散传播过程的两个互补视角：粒子性是晶格的局域更新，波动性是晶格间的相位关联。

7. 量子隧穿效应

量子隧穿——粒子穿越经典禁区的现象——在标准框架中显得神秘。在我们的框架中，隧穿可被理解为湮灭重生机制在相邻晶格间的概率性跳变。经典禁区只是世界相内电磁相窗口的能量限制，而虚空破缺的随机残余涨落允许晶格状态以有限概率在“禁止”区域重生。隧穿概率的高低，取决于该区域晶格共振的稳定程度。

8. 真空涨落与卡西米尔效应

在标准量子场论中，真空涨落被描述为虚粒子对不断产生和湮灭的过程。在我们的框架中，真空涨落对应于晶格上尚未完全共振同步的局域破缺残余——这些残余在稳定期的有序基底上表现为微弱的随机涨落。卡西米尔效应中两平行板间的吸引力，则是板间晶格涨落模式受几何限制后形成的净压力差：板外空间可以容纳更多涨落模式，由此产生了向内的净推力。

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以上合理性猜想表明，本文构建的理论框架具备广泛解释量子力学核心疑难的前景。当然，从定性猜想到定量计算之间，仍有大量工作待完成——在离散晶格上重新定义场算符、导出传播子、建立费曼规则，并在长波极限下验证标准量子场论的有效场论形式得以恢复，将是这一框架能否真正立足的关键考验。

但无论如何，我们已经迈出了第一步：从一个朴素追问出发，凭借物理直觉与反复深思，构建了一个逻辑自洽、根植于第一性原理的宇宙图景。世界本不该那么复杂——或许，它真的没有我们想象中那么复杂。